

Solution: a) There are 3 vectors in $\{v_1, v_2, v_3\}$

b) By Defn, Col A refers to the set of all linear combinations of the columns of A, i.e. $c_1 v_1 + c_2 v_2 + c_3 v_3$, where $c_1, c_2, c_3 \in \mathbb{R}$.

Let, $x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$ and $c_1 v_1 + c_2 v_2 + c_3 v_3 = x$. By Defn. of Ax,

$$[v_1 \ v_2 \ v_3] \begin{bmatrix} c_1 \\ c_2 \\ c_3 \end{bmatrix} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

$\Rightarrow Ax = x$. By Theorem 3 from Chapter 1, the matrix equation $Ax = x$ has the same solution set as the system of equations corresponding to the augmented matrix $[A \ x]$.

$$\begin{bmatrix} 2 & -3 & -4 & x_1 \\ -8 & 8 & 6 & x_2 \\ 6 & -7 & -7 & x_3 \end{bmatrix} \xrightarrow[\substack{\text{Row 2} = \text{Row 2} \\ + 4 \times \text{Row 1}}]{\substack{\text{Row 3} = \text{Row 3} \\ - 3 \times \text{Row 1}}} \begin{bmatrix} 2 & -3 & -4 & x_1 \\ 0 & -4 & -10 & 4x_1 + x_2 \\ 6 & -7 & -7 & x_3 \end{bmatrix} \xrightarrow[\substack{\text{Row 3} = \text{Row 3} \\ + \frac{1}{2} \times \text{Row 2}}]{\substack{\text{Row 3} = \text{Row 3} \\ + \frac{1}{2} \times \text{Row 2}}} \begin{bmatrix} 2 & -3 & -4 & x_1 \\ 0 & -4 & -10 & 4x_1 + x_2 \\ 0 & 0 & 0 & x_3 + \frac{x_2}{2} - x_1 \end{bmatrix}$$

It's very easy to verify that the final matrix is in echelon form. By Theorem 2 from Chapter 1, if $x_3 + \frac{x_2}{2} - x_1 \neq 0$, then the corresponding system of equations is inconsistent, i.e. x has no solutions.

Let, $x_3 + \frac{x_2}{2} - x_1 = 0$. Column 3 has no pivots. $\therefore x_3$ is a free variable.

$\therefore$ By Theorem 2 from Chapter 1, the matrix equation $Ax = x$ has infinitely many solutions when $x_3 + \frac{x_2}{2} - x_1 = 0$, i.e. $2x_3 + x_2 - 2x_1 = 0$. This is the equation of a plane, which contains infinite points.

So, Col A has infinitely many vectors $\square$

~~c) $p \in \text{Col A}$ if and only if $2x_3 + x_2 - 2x_1 = 0$~~ By part (b), we know that

$p \in \text{Col A}$ if and only if $2x_3 + x_2 - 2x_1 = 0$. $p = \begin{bmatrix} 6 \\ -10 \\ 11 \end{bmatrix}$

$$2 \times 11 + (-10) - 2 \times 6 \Rightarrow 22 - 10 - 12 = 0.$$

$\therefore p \in \text{Col A} \square$